INTERNSHIP REPORT

ON

GENAI-POWERED DATA ANALYTICS AND DELINQUENCY RISK PROFILING

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

SUSHIL KUMAR

Roll Number of Student: 25SCS1003004474

Batch: 2026-27

CANDIDATE’S DECLARATION

I, SUSHIL KUMAR, do hereby declare that the internship report titled “GenAI-Powered Data Analytics and Delinquency Risk Profiling” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. The report presents the work completed as part of the GenAI Powered Data Analytics Job Simulation issued through Forage. All references used in preparing this report have been acknowledged. I affirm that this report is my own academic work and has not been submitted to any other institute for any degree or diploma requirement. I shall be responsible for any copyright violation arising from material submitted by me.

Signature: Sushil kumar

Student Name: SUSHIL KUMAR

Roll No.:-25SCS1003004474

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported and guided me during the internship and the preparation of this report. I am thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing an academic environment in which practical learning and professional development are encouraged.

I am also grateful to Forage for providing the GenAI Powered Data Analytics Job Simulation. The simulation enabled me to work through practical data-analytics activities involving exploratory data analysis, risk profiling, AI-based delinquency prediction, business reporting, data storytelling, and an AI-driven collections strategy.

I would also like to thank my faculty members, mentors, classmates, and family for their guidance, encouragement, and continuous support. Their feedback helped me understand how technical analysis can be converted into clear and useful business recommendations.

Finally, I acknowledge the learning experience gained through this internship-style simulation, which strengthened my understanding of data analytics and its application to business decision-making.

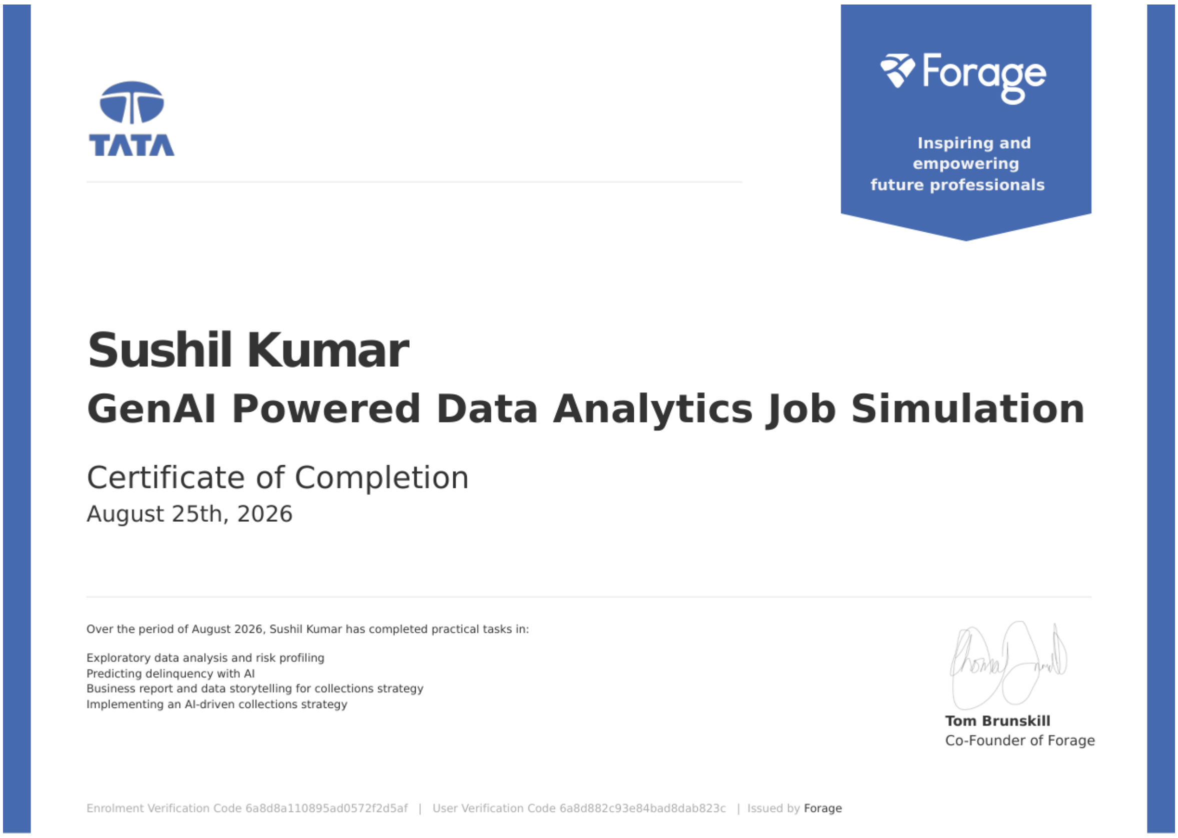
Signature: ______________________________

Student Name: SUSHIL KUMAR

Roll No.:-25SCS1003004474

INTERNSHIP COMPLETION CERTIFICATE

A copy of the internship/job-simulation completion certificate issued by Forage is attached below. The certificate confirms that Sushil Kumar completed the "GenAI Powered Data Analytics Job Simulation" in August 2026 and completed practical tasks in exploratory data analysis and risk profiling, predicting delinquency with AI, business reporting and data storytelling for collections strategy, and implementing an AI-driven collections strategy.



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Forage

Inspiring and empowering future professionals

Sushil Kumar

GenAI Powered Data Analytics Job Simulation

Certificate of Completion

August 25th, 2026

Over the period of August 2026, Sushil Kumar has completed practical tasks in:

Exploratory data analysis and risk profiling
Predicting delinquency with AI
Business report and data storytelling for collections strategy
Implementing an AI-driven collections strategy

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a8d8a110895ad0572f2d5af | User Verification Code 6a8d882c93e84bad8dab823c | Issued by Forage

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4. PROJECT DESCRIPTION

The internship work was completed through the GenAI Powered Data Analytics Job Simulation issued by Forage. The project focused on applying data-analytics and AI concepts to a financial-services collections scenario. The certificate records four practical areas: exploratory data analysis and risk profiling; predicting delinquency with AI; business reporting and data storytelling for collections strategy; and implementing an AI-driven collections strategy.

The work was structured as a progression from understanding the data to converting analytical findings into decisions. First, the available customer information was examined for structure, quality, patterns, and potential risk indicators. Next, a predictive framework was considered for identifying customers who may be at higher risk of delinquency. The analytical results were then translated into a business report and a practical collections strategy.

This project therefore connects three important capabilities: data preparation, predictive analytics, and business communication. It demonstrates how an analyst can move from raw customer data to risk insights and then to actionable recommendations without treating a model output as a replacement for human judgement.

4.1 INTRODUCTION

Data analytics is the process of collecting, cleaning, examining, modelling, and interpreting data to support informed decisions. In financial services, analytics can help organizations identify patterns in customer behaviour, understand repayment risk, prioritize interventions, and allocate operational resources more effectively.

Generative AI and conventional machine-learning methods can support different parts of an analytics workflow. Generative AI can assist with summarizing findings, organizing business narratives, and translating technical results into stakeholder-friendly language. Predictive models can estimate the likelihood of an outcome from historical variables. Exploratory data analysis provides the foundation by revealing data-quality issues, distributions, relationships, and unusual observations before predictive modelling begins.

The internship project applied these ideas to a delinquency-risk and collections use case. The goal was not simply to produce a prediction, but to build a logical chain from data understanding to risk profiling, prediction, business reporting, and an intervention strategy.

The certificate supplied with this report confirms completion of the practical simulation in August 2026. It specifically identifies exploratory data analysis and risk profiling, AI-based delinquency prediction, business reporting/data storytelling, and an AI-driven collections strategy as the completed practical tasks.

4.2 ORGANIZATION PROFILE

Forage is a career-development platform that provides virtual job simulations designed to give learners exposure to realistic work tasks. Its official website describes job simulations as self-paced experiences in which learners complete tasks that replicate real work, compare their work with model answers, and earn a certificate on completion. Forage offers simulations across areas such as data and analytics, software engineering, cybersecurity, consulting, and banking and finance.

The present internship-style experience was completed through Forage and was titled "GenAI Powered Data Analytics Job Simulation." The completion certificate identifies Tom Brunskill, Co-Founder of Forage, and states that the simulation was completed by Sushil Kumar during August 2026.

The learning model is particularly relevant to computer-science students because it emphasizes practical application rather than only theoretical study. A job simulation provides a controlled environment in which students can practise analysis, communication, structured problem solving, and professional reporting.

For this report, Forage is treated as the platform through which the simulation was delivered. The project scenario itself is treated as a case-based learning exercise rather than as evidence of employment in a physical office or production organization.

4.3 PROBLEM STATEMENT

The central problem addressed in the project is how data and AI can be used to identify customers who may be at increased risk of payment delinquency and how those analytical insights can be translated into an effective collections strategy.

A collections team may have access to many customer-level variables, but raw data alone does not provide a clear prioritization of cases. The challenge is to understand the available information, identify meaningful risk indicators, develop a structured prediction approach, communicate the findings clearly, and recommend interventions that are proportionate to the identified risk.

The problem can therefore be expressed as follows: given customer and financial attributes, determine which patterns are associated with higher delinquency risk, construct a transparent predictive framework, and convert the resulting risk segments into practical and responsible collections actions.

A key consideration is that predictive analytics should support decision-making rather than automatically determine a customer's treatment. Model outputs need validation, monitoring, and appropriate human review, especially when decisions can affect individuals.

4.4 PROJECT OBJECTIVES

The major objectives of the internship project were:

- To understand the structure and quality of a customer-level delinquency dataset.
- To perform exploratory data analysis and risk profiling before predictive modelling.
- To identify variables and customer segments that may be associated with delinquency risk.
- To develop a structured AI/predictive modelling approach for delinquency prediction.
- To explain model inputs and outputs in a way that is understandable to non-technical stakeholders.
- To convert analytical findings into a business report and data-storytelling narrative.
- To design an AI-driven collections strategy that prioritizes appropriate intervention.
- To strengthen practical skills in data analytics, predictive reasoning, documentation, and business communication.

4.5 SCOPE OF THE PROJECT

The scope of the project covers the complete analytical lifecycle represented in the job simulation: data understanding, exploratory analysis, risk profiling, predictive modelling, reporting, and strategy formulation.

Within scope are:

- Reviewing customer-level data and identifying important variables.
- Checking data quality, missing values, inconsistent values, and distributions.

- Using descriptive and visual analysis to understand patterns.
- Defining a predictive target for delinquency and considering suitable model families.
- Comparing a simpler, interpretable modelling approach with a more complex alternative at a conceptual level.
- Preparing business-oriented summaries of risk factors and customer segments.
- Designing a prioritization framework for collections interventions.
- Documenting the internship learning and attaching the completion certificate.

Out of scope are live deployment to a banking/collections production system, real customer decisions, direct access to confidential financial accounts, and claims about actual business performance after deployment. The work is an internship/job-simulation case study and should be interpreted accordingly.

4.6 TECHNOLOGIES AND TOOLS USED

The project uses a data-analytics workflow in which spreadsheet-style data handling, statistical exploration, predictive modelling concepts, visualization, and generative-AI-assisted communication can be combined. The following tools and technologies are relevant to the completed practical work and the analytical workflow documented in this report.

Tool / Technology	Purpose	Application in Project
Python	Data analysis and modelling	Data preparation, exploratory analysis and predictive workflow
Pandas	Tabular data processing	Cleaning, filtering, grouping and summarizing customer data
NumPy	Numerical computation	Numerical transformations and analytical calculations
Matplotlib	Data visualization	Charts for distributions, comparisons and risk patterns
Scikit-learn	Machine learning	Structured framework for classification and model evaluation
Decision Tree / Logistic Regression	Predictive modelling	Interpretable baseline approaches for delinquency-risk prediction
Generative AI	Analysis and communication support	Summarizing findings, structuring reports and supporting business storytelling
Microsoft Word	Documentation	Preparation of the internship report and stakeholder-ready documentation

4.7 SYSTEM ARCHITECTURE

The proposed analytical architecture follows a sequential pipeline. Customer-level data first enters the data-preparation layer. After quality checks and exploratory analysis, relevant features are transformed into a modelling dataset. A predictive model estimates delinquency risk, after which the risk outputs are grouped into actionable segments. Finally, the analytical results are communicated through a business report and collections strategy.

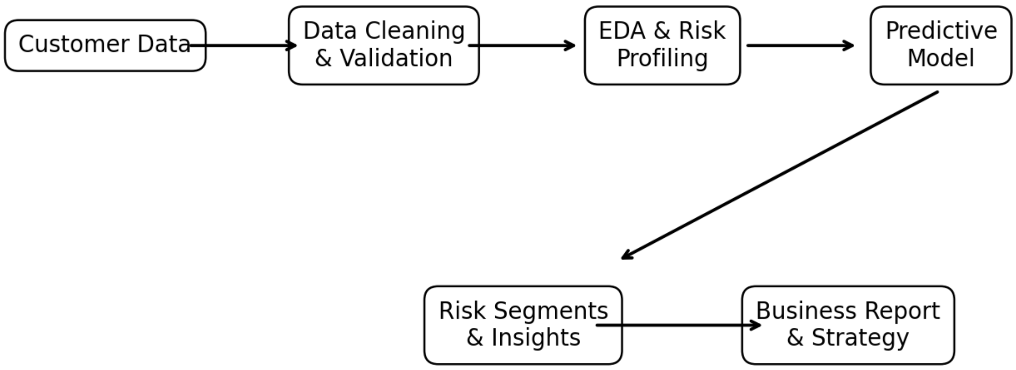

Figure 4.1: Proposed GenAI-supported delinquency analytics architecture

The architecture separates analytical stages so that each stage can be checked independently. Data quality checks occur before modelling; exploratory analysis is used to understand the variables; the model produces a risk estimate; and the final stage translates risk information into business actions. This separation improves transparency and makes the workflow easier to validate and maintain.

4.8 METHODOLOGY

The methodology adopted for the project is a structured analytics lifecycle. It starts with understanding the business problem and dataset, proceeds through exploratory analysis and predictive modelling, and finishes with communication and intervention planning.

4.8.1 Step 1 – Business and Dataset Understanding

The first step is to understand what the prediction is expected to achieve. In a delinquency context, the target is whether a customer is likely to become delinquent within the defined observation period. The dataset is then reviewed for its rows, columns, data types, categorical variables, numerical variables, identifiers, and potential target leakage.

This stage prevents modelling from beginning with an unclear objective. It also helps distinguish variables that are suitable predictors from fields that should not be used because they are identifiers, post-outcome information, or otherwise inappropriate for prediction.

4.8.2 Step 2 – Data Quality and Pre-processing

Data preparation includes checking missing values, duplicate records, invalid ranges, inconsistent categories, unusual numerical values, and the distribution of the target variable. Numerical variables may require appropriate scaling for some model families, while categorical variables may require encoding.

A reproducible preprocessing pipeline is preferred because the same transformations must be applied consistently during model training and future prediction. Data quality findings should also be documented rather than silently changed.

4.8.3 Step 3 – Exploratory Data Analysis

Exploratory data analysis is used to understand the structure and patterns in the dataset. Descriptive statistics can reveal central tendency and spread, while frequency tables can show category distributions. Visualizations such as histograms, box plots, bar charts, and correlation views can help identify patterns and potential relationships.

For delinquency analysis, the analyst should compare the target outcome across relevant customer and financial characteristics. Examples include payment behaviour, debt burden, income-related variables, account activity, and other variables available in the dataset. The purpose is to identify useful signals while avoiding unsupported conclusions about causality.

4.8.4 Step 4 – Risk Profiling

Risk profiling converts exploratory findings into understandable customer segments. Instead of treating all customers as equally risky, the analysis can identify groups with comparatively higher or lower predicted risk. Segment definitions should be based on evidence in the data and should be understandable to the collections team.

Risk profiling also supports resource allocation. Higher-risk segments may require earlier or more frequent review, while lower-risk segments may be handled through lighter-touch communication. The exact operational treatment should remain subject to organizational policy and human review.

4.8.5 Step 5 – Predictive Modelling Framework

The project considers a binary classification problem: predicting whether an observation belongs to a delinquent or non-delinquent class. A transparent baseline such as logistic regression can provide a useful benchmark because its coefficients can be interpreted directionally. A decision tree can provide intuitive rule-based splits and may capture non-linear relationships.

For a more complex approach, ensemble tree methods could be considered if additional performance is required. However, model complexity should be balanced against interpretability, monitoring requirements, data quality, and the need to explain predictions to business stakeholders.

A practical modelling workflow is: split data into training and validation sets; fit preprocessing and model components on training data; evaluate on held-out data; inspect class imbalance; select appropriate metrics; and review errors before considering deployment.

4.8.6 Step 6 – Model Evaluation

Model evaluation should not rely only on accuracy, especially when delinquency outcomes are imbalanced. Precision, recall, F1-score, ROC-AUC, and a confusion matrix can provide a broader view of model performance. In a collections setting, recall for the high-risk class may be important, but excessive false positives can also waste resources or lead to inappropriate customer treatment.

The final model should therefore be assessed from both technical and business perspectives. A model with slightly lower predictive performance may be preferable if it is substantially easier to explain, monitor, and govern.

4.8.7 Step 7 – Business Reporting and Data Storytelling

After analysis, the findings are translated into a concise business narrative. A stakeholder report should answer three questions: what was found, why it matters, and what should be done next. Technical metrics are included only where they help decision-makers understand confidence, limitations, or trade-offs.

The report should highlight the most important risk factors, describe high-risk segments, explain the recommended intervention framework, and identify limitations. Generative AI can assist with organizing and refining the narrative, but the analyst remains responsible for checking that the final claims are supported by the underlying analysis.

4.8.8 Step 8 – AI-Driven Collections Strategy

The final stage links risk predictions to intervention tiers. A simple framework can use three broad categories: lower risk, moderate risk, and higher risk. Each tier can have a different communication intensity and review priority.

Risk Tier	Analytical Meaning	Suggested Focus	Governance Consideration
Lower	Lower predicted probability of delinquency	Routine reminders and standard service	Avoid unnecessary intervention
Moderate	Meaningful but not highest predicted risk	Timely reminders, monitoring and supportive outreach	Review customer context before escalation
Higher	Highest predicted probability among modelled groups	Earlier review and prioritized outreach	Human review and policy compliance are essential

4.8.9 Step 9 – Monitoring and Continuous Improvement

A deployed predictive system should be monitored over time because customer behaviour, economic conditions, policies, and data distributions can change. Monitoring should cover model performance, data drift, missingness, segment volumes, intervention outcomes, and potential differences in error rates across relevant groups.

Feedback from collections teams can also be incorporated into future model improvements. The objective is not simply to maximize prediction accuracy but to create a useful, responsible, and measurable decision-support system.

4.9 EXPECTED OUTCOMES

The expected outcomes of the project are both technical and professional. On the technical side, the project develops a repeatable workflow for examining customer data, profiling risk, and structuring a delinquency prediction model. On the business side, it demonstrates how analytical results can be converted into recommendations for collections prioritization.

- A clearer understanding of customer-data quality and structure.
- Identification of important patterns and potential delinquency risk indicators.
- A transparent predictive-modelling framework suitable for further validation.
- Risk segmentation that can support prioritization of collections activity.
- A stakeholder-ready narrative explaining findings, implications, and recommendations.

- An AI-driven collections strategy that combines analytical prioritization with human oversight.
- Improved skills in data storytelling, report writing, analytical reasoning, and professional communication.

The certificate confirms completion of the practical tasks, while the report documents the analytical framework and learning outcomes associated with those tasks. Because this was a job simulation rather than a live production deployment, no claim is made that the proposed strategy generated actual collections improvements.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATION PROOF

The internship completion certificate issued by Forage is attached in this report. It identifies the learner as Sushil Kumar and confirms completion of the GenAI Powered Data Analytics Job Simulation in August 2026. The certificate lists the practical tasks completed: exploratory data analysis and risk profiling; predicting delinquency with AI; business report and data storytelling for collections strategy; and implementing an AI-driven collections strategy.

The certificate also identifies Forage as the issuing organization and includes verification information. The attached certificate should be retained as the primary proof of completion.

No separate communication email or organizational mail proof was supplied with the materials available for this report. Therefore, no unsupported email correspondence has been created or represented as genuine. If the university requires additional communication proof, the student should insert the original email/screenshot in this section before final submission.

5. BIBLIOGRAPHY / REFERENCES

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